# Walk calibration of BTL modules in testbeam:

## Via Synchronization of MCP scope data with FEBd

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Discussion draft; not approved for public release.

## Purpose: provide a reliable time reference for BTL analyses

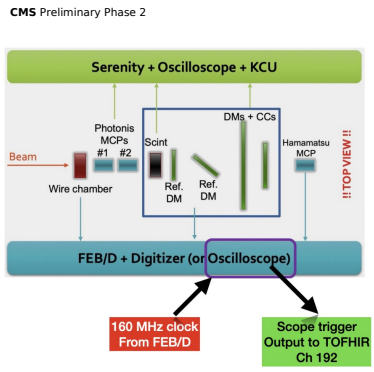
Several BTL test-beam analyses need a reliable particle-arrival time reference.

For a subset of the data, a digital oscilloscope recorded MCP-PMT signals in parallel with the BTL readout.

**Approach:** synchronize the two streams, use the MCP to calibrate upstream reference bars, then use their corrected times as the reference for downstream timing.

**Final check:** the time difference between downstream and reference bars before and after walk correction.

# Parallel MCP and BTL streams are linked through FEBd

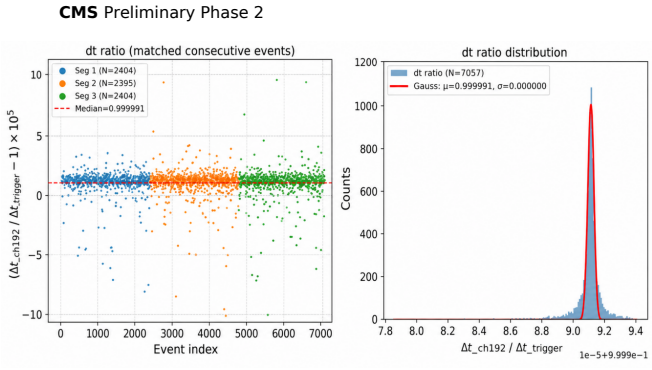



**Two readout streams**
The scope records the MCP-PMT pulse. FEBd records the BTL SiPM signals.

**Synchronization link**
The scope trigger also reaches TOFHIR channel 192 in FEBd. This link connects independently clocked records of the same particle.

September 2025 beam test. The highlighted lower path provides the reference-bar data. This FEBd-based synchronization is specific to the test-beam setup, not the final HL-LHC synchronization scheme.

# Events are correlated using the intervals between triggers

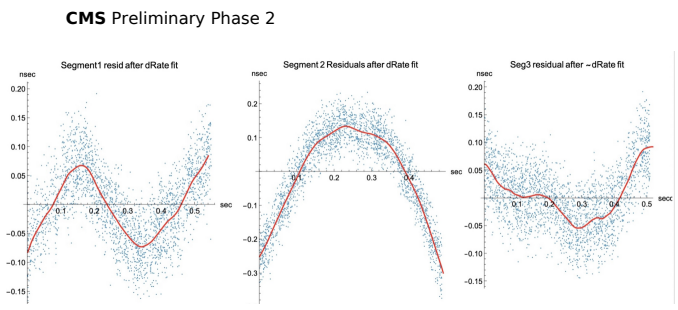



1. Compare consecutive trigger intervals in channel 192 and the scope.
2. Find consistent interval patterns.
3. Keep only unique event matches.

Consistent trigger patterns identify coincident events in the two streams.
Event matching alone does not remove relative clock drift.

The correlated intervals and their ratio test event matching, not detector resolution. GitHub: trigger matching.

# The matched clocks show low-frequency relative drift



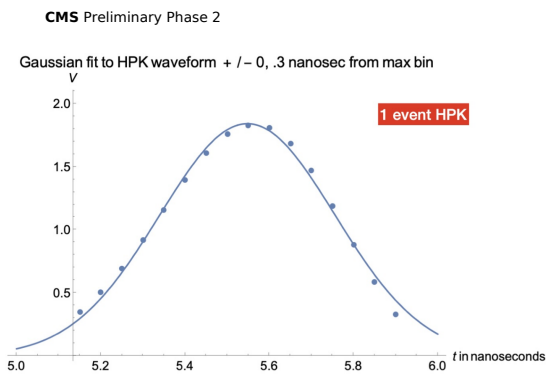

## Correct the timing coordinate

1. Split data at clock resets.
2. Fit a line between matched trigger times in each segment.
3. Fit the remaining drift with LOWESS.

Add the drift to the line prediction to obtain $g(u)$, the expected TOFHIR trigger time.

Points show residuals before LOWESS; red curves describe smooth drift. Reject groups with pre-LOWESS residual scale $1.4826\,\mathrm{MAD} > 1$ ns. GitHub: clock reconstruction.

# After synchronization, the MCP defines the calibration time




Fit the MCP waveform to estimate its pulse time.
Within the scope waveform, define

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}} - t_{\mathrm{trigger}}.$$

For SiPM channel $i$, use

$$\Delta t_i = t_i - g(u) - \delta_{\mathrm{MCP}}.$$

The charge-dependent part of this difference is the walk.

The waveform and fitted template determine the MCP pulse time. $u$ is the scope-trigger timestamp. A fixed cable/geometry offset is irrelevant to the shape; clock-rate conversion over the short waveform interval is neglected.

# Measure the upstream channels' time walk versus charge

**Separate threshold settings before selection and fitting.**

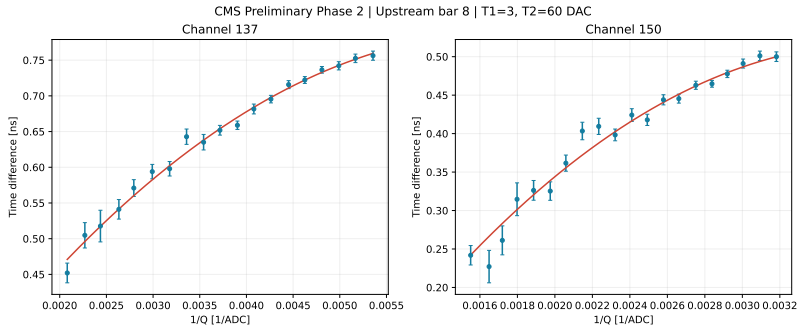
1. In run 4399, require exactly five TOFHIR hits and a unique MCP match: upstream channels 137 and 150, downstream channels 234 and 243, and channel 192.
2. Within each setting, require clock quality and finite positive charge, $\Delta I$ and threshold-rise time. Keep charge above the spectrum's left half-maximum boundary.
3. Form 20 uniform $1/Q$ bins. Clip timing outliers at $5(1.4826\,\mathrm{MAD})$ from each bin's median; require at least five entries.
4. Fit the bin means with weights $1/\mathrm{SEM}^2$:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

$Q$ is the measured ADC charge proxy, not energy in MeV. SEM denotes the standard error of the mean.

Upstream bar 8 and downstream bar 3. Combine files only at identical scan settings. GitHub: selected reference sample.

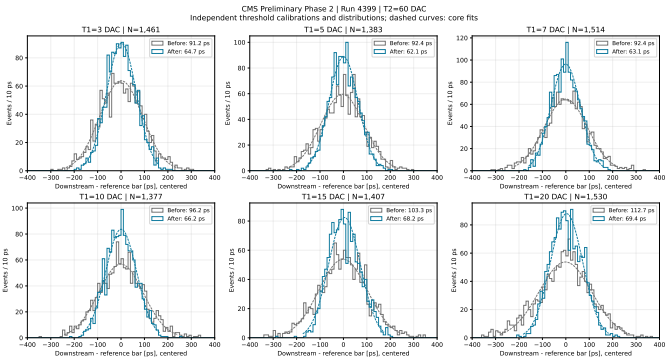
# Walk-corrected upstream bars provide the time reference




Correct each end: $t_i^{\mathrm{corr}} = t_i - W_i(Q_i)$. Use the bar average: $t_{\mathrm{ref}} = (t_L^{\mathrm{corr}} + t_R^{\mathrm{corr}})/2$.

Upstream bar 8, lowest scanned setting: $T_1 = 3$, $T_2 = 60$ (DAC settings). Points show profile means with SEM; curves are SEM-weighted quadratics in $1/Q$. The $+26$ ns display offset does not change the shape. Separate fits are made for every setting and all four channels.

# Final result: downstream-bar minus reference-bar timing




Difference of bar-average times: downstream bar 3 minus upstream bar 8. Corrected core widths: 62–69 ps. For independent timing fluctuations, $\sigma_{\Delta t} = \sqrt{\sigma_{\mathrm{up}}^2 + \sigma_{\mathrm{down}}^2} \approx 65\,\mathrm{ps}$; individual widths are not separately measured. Thresholds are fitted separately with identical before/after events. Display-only median centering; no per-file offsets. Dashed curves: truncated-normal fits within median $\pm 2.5(1.4826\,\mathrm{MAD})$. In-sample validation.

# References


[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[5] S. White, Nov. 2025 BTL TB Analysis mtg presentation, `https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf`

[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

# Method for Walk Correction using Leading Edge Slope:

## Derived from dual threshold feature of TOFHIR2C

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Discussion draft; not approved for public release.

## Purpose: rapid, repeatable walk calibration at HL-LHC

BTL needs frequent recalibration as detector response evolves. For about 330 000 channels, a consistent physical parametrization matters.

**Goal:** predict each channel's walk correction from simple internal charge and dual-threshold measurements, without a dedicated $t_0$ reference for every calibration.

TOFHIR2C measures the leading-edge slope. This gives a common correction form and internal observables with which to predict its amplitude, AWC. The MCP study benchmarks this approach; transfer to new data and operating conditions is the next test.

BTL and TOFHIR2C: Refs. [3, 7]; dual-threshold motivation and bench study: Ref. [5].

## Charge-based calibration works, but its coefficients vary

The MCP reference note obtains good walk descriptions with a quadratic in inverse charge [6]:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

| Bar | Channel | $a_i$ [ns] | $b_i$ [ns ADC] | $c_i$ [ns ADC$^2$] |
|---|---|---|---|---|
| 7 | 133 | 0.0796 | 238.3 | $-2.156 \times 10^4$ |
| 7 | 154 | $-0.5155$ | 422.6 | $-5.583 \times 10^4$ |
| 8 | 137 | 0.1218 | 198.5 | $-1.523 \times 10^4$ |
| 8 | 150 | $-0.2292$ | 388.9 | $-5.132 \times 10^4$ |

The shape coefficients vary substantially even among four channels. Use pulse slope to seek a more consistent physical description.

Original reference coefficients, also reproduced in Table 1 of the detector note. $Q$ is the ADC charge proxy; $a_i$ includes an arbitrary timing offset. Coefficient variation alone is not a measure of calibration precision.

## TOFHIR2C measures the leading-edge slope with two thresholds

Larger pulses cross a fixed threshold earlier: the timestamp depends on amplitude. Two thresholds also measure how fast the pulse rises [7, 5].

$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

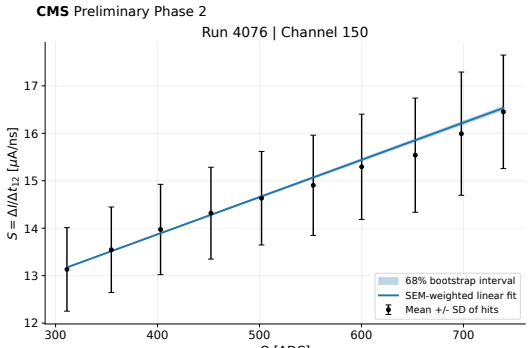
Locally, an approximately linear leading edge gives

$$\Delta t \simeq \frac{\Delta I}{S}.$$

**Physical model:** describe amplitude walk with an inverse slope, rather than an independent polynomial shape for every channel.

$S$ is a finite-interval slope proxy, not the instantaneous derivative. Run 4076 uses $\Delta I = 7.8125\,\mu$A from the implemented threshold conversion. The fitted AWC need not equal this threshold separation. GitHub: timing and threshold reconstruction.

# Measure the charge-to-slope mapping using internal hits




Select both ends of one bar; leave the other module unconstrained. Use physical threshold crossings and the peak-right charge region. Fit ten profile means with SEM weights:

$$\overline{S}_i(Q) = c_{0,i} + c_{1,i} Q.$$

No MCP time enters this fit.

Channel 150, run 4076: 19 312 selected hits. Points are mean slopes; bars show hit SD, not uncertainty on the mean. The band is the conditional 68% interval from 1000 hit-bootstrap refits with fixed bins. The profile averages the noisy single-hit slope measurements.

## One amplitude describes walk once the slope is calibrated

$$W_i(Q) = \frac{A_i}{c_{0,i} + c_{1,i} Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_i(Q).$$

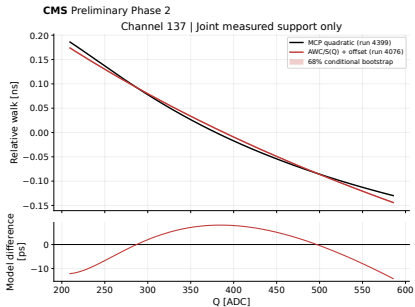
- Internal hits determine $c_{0,i}$ and $c_{1,i}$, the charge-to-slope mapping.
- One amplitude $A_i$ (AWC, in $\mu$A) then sets the walk magnitude.
- AWC is constant per channel and configuration; the correction varies with $Q$.
- Channel timing offsets are a separate calibration.

**The remaining task is to predict AWC from internal observables.** Use the MCP-calibrated channels to develop and test that prediction.

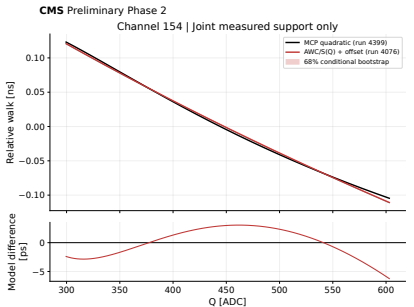
Apply the model within its calibrated charge range, with $c_{0,i} + c_{1,i} Q > 0$. The slope fit alone does not fix AWC. The common functional form follows the dual-threshold model in Ref. [5].

# The single-amplitude model closely follows the MCP curves

## Channel 137

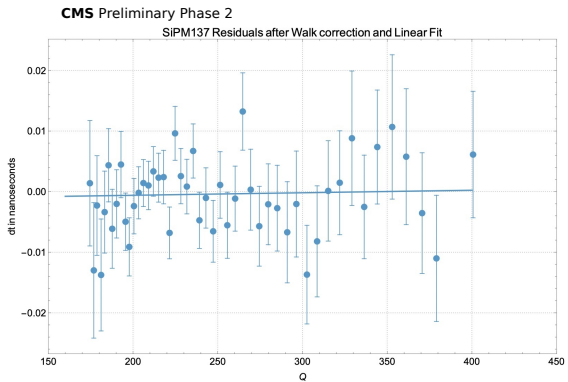



## Channel 154




Black: MCP quadratic (run 4399); red: inverse-slope model (run 4076). AWC and an offset are fitted over charge intervals populated in both runs, which contain different events. Channel 137 has 6.80 ps RMS curve mismatch. Lower panels show curve differences, not event resolution. Bands propagate slope-fit statistics with the MCP curve fixed. This tests the functional form, before predicting AWC.

# Residual charge dependence after the AWC correction




**Channel 137**

Inspect the timing residual versus charge after the inverse-slope correction.

$0.01\,\mathrm{ns} = 10\,\mathrm{ps}$.

The residual linear trend is nearly flat over the displayed range.

This is not a single-hit timing-resolution measurement.

Original detector note, Figure 5: residual MCP-referenced timing, $t_{\mathrm{thr1}} - t_0$, after applying the AWC-type walk correction. Points and error bars are reproduced unchanged; the line is the original residual linear fit. This historical figure complements the curve-to-curve comparison on the preceding slide.

# References


[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

[2] "PID performance of the ALICE-TOF detector at Run 2", N. Jacazio in Proceedings of LHCP2018, https://pos.sissa.it/321/232/.

[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[4] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[5] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

[7] E. Albuquerque et al., "TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector", arxiv:2404.01208

[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

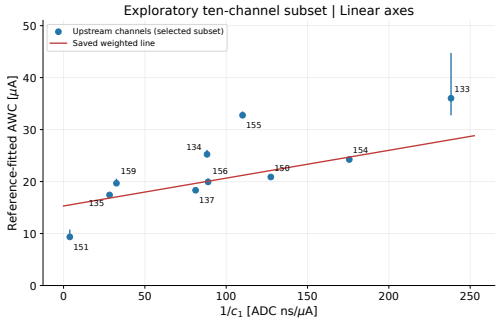
[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) https://doi.org/10.1016/0168-9002(94)01239-3.

[10] This will arise in LGAD that are not thinned to remove the effect (J. Vavra, private communication). See also PICOSEC, "Fast Timing with MicroPattern Gas Detectors", where charge in the electron peak is captured-ignoring the ion tail..

[11] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

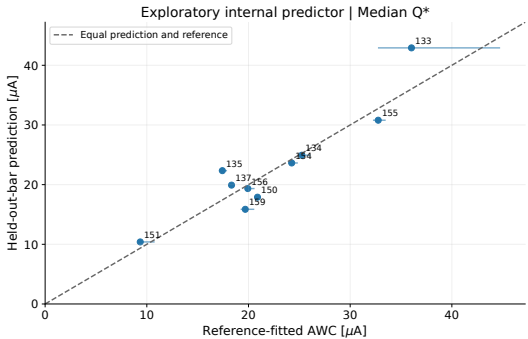
[12] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.

# Backup: empirical AWC trend with inverse slope coefficient




The final observable of the original detector note: AWC versus $1/c_1$, the inverse slope-to-charge coefficient. Each point is one channel, not a charge bin. AWC is benchmarked against MCP; $c_1$ uses internal hits. The ten-channel trend motivates prediction, but the line is not a precision calibration ($\chi^2/\mathrm{ndf} = 538.6/8$). Bars: slope-only 16–84% bootstrap intervals; reference errors and correlations are omitted. More independent data are needed.

# Backup: a shared fractional-charge response hypothesis




Hypothesis: common $K \simeq -dW/d\ln Q$ at $Q_*$, not a derived law.

$$A_i = K\,\frac{\overline{S}_i(Q_*)^2}{c_{1,i}\,Q_*}.$$

$Q_*$: selected-charge median per channel.
Learn $K$ on other bars; predict the excluded bar.
RMS error: 3.22 $\mu$A; fractional RMS: 14.1%.

Ten channels, six held-out bars. $K$ uses other bars' MCP-calibrated AWC. Horizontal bars: reference-AWC intervals. Empirical hypothesis only: full prediction uncertainties, correlations and transfer to new configurations remain unvalidated.